

Sacha Arseneault

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EDUCATION

University of Ottawa

BASc in Software Engineering, Co-op

Ottawa, ON

GPA: 3.9/4.0 | Expected: Dec 2026

EXPERIENCE

Microsoft AI

Applied AI/ML Intern — Copilot

Mountain View, CA

June 2026 - Aug 2026

- Supporting Copilot's agent stack serving **+100M monthly users**, across harness engineering and response quality.
 - Architected a new evidence bundling pipeline to reinforce live evaluations against long-running agentic coding sessions, powering runtime-orchestration flywheels and **reducing eval runtimes by over 40%**.
 - Improved first-turn **KV cache hit rate** on Copilot's GPT-based inferencing from 0.9% to 90% by introducing cross-conversational prefix routing in GPU load balancing.
 - Demo'd a proposal on how autonomous agents should handle self-directed continuity to VP-level suite.

Tesla

Software Engineer Intern — Robotaxi

Palo Alto, CA

May 2025 - Aug 2025

- Developed **infrastructure** and executed **live validation** supporting Tesla's official **robotaxi** launch. (June '25)
 - Built an integration test suite validating telemetry received from 6 different pipelines during **real robotaxi rides**, reducing full-system verification from 15 to 2 minutes. (Go, Docker, Bash, Kafka)
 - Conducted **live QA across 200+ physical rides**, identifying dispatch and rider experience issues and coordinating resolution with cross-functional teams ahead of Early Access.
 - Added pre-merge detection for versioning-related schema mismatches (Redis), reducing deployment regressions to zero.
 - Configured post-merge **canary workflows** executing end-to-end validation **across the complete rider journey**—from ride hailing through passenger drop-off—by orchestrating mobile emulators and simulated car fleets.
 - Implemented **fault-tolerant parsing** across all telemetry pipelines, recovering 100% of valid fields in bad payloads.

Shopify

Software Engineer Intern — Stores

Ottawa, ON

Jan 2025 - Apr 2025

- Increased **robustness** and **resiliency** in Shopify's store-creation process and developer platform.
 - Resolved a 1/1000 **race condition** between asynchronous Kafka events and GraphQL mutations affecting over 2 million store creations annually, preventing duplicate records across 3 MySQL databases.
 - Implemented a **self-healing timeout job** for expired store-creation requests, achieving 100% recovery and eliminating manual intervention during platform outages. (Ruby, Rails)

LEADERSHIP

IEEE uOttawa - Webmaster

Sep 2022 – Apr 2026

- Led a team of 10 developers in building 4 **open-source web-apps** driving 5,000+ visits and 25,000+ impressions per month, tracked by the successful review and merge of 75+ pull requests.
- Hosted 2 workshops (**Git/React**) and 3 review sessions (**Data Structures and Algorithms**) with 50+ attendees each.

PROJECTS

Concurrency Debugger for Go | Golang, Websocket, GinkGo

2026

- A multi-platform concurrency debugger, enabling visualization of goroutines, channels, and synchronization.
- Co-designed and built the **full system architecture in Go**, including the core debugger engine, a session-based DAP server, and an interactive client for session control, execution, and breakpoint management.
- Developed **low-level debugging backends** for Linux (amd64) and macOS Silicon (darwin, arm64), leveraging OS-specific process control and metadata to **facilitate runtime inspection**.

Eco Rewards App | Remix, TypeScript, Express.js, PostgreSQL, ShadCn, Auth0

2024

- A full-stack points-tracking app rewarding users for sustainable day-to-day actions.
- Won 1st Place** out of 842 participants in the top sponsor track at **Hack the North**, Canada's largest hackathon.

TECHNICAL SKILLS

Languages: Go, Python, TypeScript, Java, SQL, Ruby

Frameworks/Libraries: Express, Next, React, Redux, PostgreSQL, Rails, Supabase, FastAPI, Scikit-learn, Pandas, Torch

Tools: Git, Docker, Linux, Bash, Prometheus, Grafana, OpenTelemetry, Kafka, Redis, GraphQL, gRPC, Protobuf